Set Theory for Computer Science

Week 2: Relations

Lists and product sets

Relations and their representations

Inverse and composite relations

Properties of binary relations

Definition and notation of lists

A **list** is an ordered sequence (an *array*) of elements

Examples

$$\begin{aligned} \mathit{DigitList} &:= \langle 0, 1, 2, \ldots, 9 \rangle \\ \mathit{LetterList} &:= \langle \mathrm{a}, \mathrm{b}, \mathrm{c}, \ldots, \mathrm{z} \rangle \\ \mathit{MixedList} &:= \langle \mathrm{t}, \mathrm{h}, \mathrm{u}, \ \ , \mathrm{f}, \mathrm{e}, \mathrm{b}, \ \ , 1, 1 \rangle \end{aligned}$$

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$$\mathit{LetterList} := \langle \mathrm{a}, \mathrm{b}, \mathrm{c}, \ldots, \mathrm{z}\rangle$$
$$\mathit{MixedList} := \langle \mathrm{t}, \mathrm{h}, \mathrm{u}, \ \ , \mathrm{f}, \mathrm{e}, \mathrm{b}, \ \ , 1, 1\rangle$$

- elements in a list have **positions** $\langle 1, 2\rangle \neq \langle 2, 1\rangle$
- finite lists have a **length** $\langle 1, 2\rangle \neq \langle 1, 2, 1\rangle$

t	h	u		f	e	b		1	1
↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
1	2	3	4	5	6	7	8	9	10

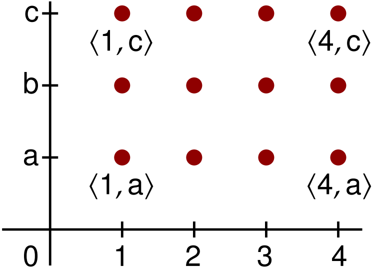
Product sets

Cartesian product of two sets

$A \times B := \{\langle a, b\rangle : a \in A \text{ and } b \in B\}$ set of **pairs**

$A \times A =: A^2$ (lists of length 2)

Example



Cartesian product
$\{1, 2, 3, 4\} \times \{a, b, c\}$

Product sets

Cartesian product of three sets

$A \times B \times C := \{\langle a, b, c\rangle : a \in A, b \in B, c \in C\}$ set of **triples**

$A \times A \times A =: A^3$ (lists of length 3)

Cartesian product of n sets

$A_1 \times \cdots \times A_n := \{\langle a_1, \ldots, a_n\rangle : a_i \in A_i \text{ for each } i\}$ (lists of length n)

$\underbrace{A \times \cdots \times A}_{n \text{ factors}} =: A^n$

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Examples

$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$	the real plane	e.g. $\langle 1, \pi\rangle \in \mathbb{R}^2$
$\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$	3d space	e.g. $\langle 0, \sqrt{2}, -1\rangle \in \mathbb{R}^3$

Size of a product set

The product rule

$\#(A \times B) = \#A \cdot \#B$

$\#(A_1 \times \cdots \times A_n) = \#A_1 \cdot \#A_2 \cdots \#A_n \qquad \Rightarrow \#(A^n) = (\#A)^n$

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Examples

$MonthNums := \{01, 02, 03, \ldots, 12\}$

$DayNums := \{01, 02, 03, \ldots, 31\} \qquad \langle 03, 10 \rangle$ 3rd of October

$Dates := DayNums \times MonthNums \qquad \Rightarrow \#Dates = 31 \cdot 12 = 372$

$Alphabet := \{\mathrm{a}, \mathrm{b}, \ldots, \mathrm{z}\}$

$\Rightarrow \#(Alphabet^4) = 26^4 = 456\,976$ number of 4-letter words

Exercise 1

Let $A := \{1, 2\}$ and $B := \{2, 3\}$. Describe the sets $A \times B$ and $B \times A$ explicitly by listing their elements. Is $A \times B$ equal to $B \times A$?

Exercise 2

The number plate of a car consists of a digit, then three letters, and then two digits. Which product of the sets *Digits* and *Letters* lists all possible number plates? How many number plates are there?

Lists and product sets

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Inverse and composite relations

Properties of binary relations

Relations

A **relation** R of **type** $A_1 \times \cdots \times A_n$ is a subset of $A_1 \times \cdots \times A_n$

Example

Thu, Feb 11, 2016

Time	Platform	Destination	Train
11:00	4	Nijmegen	Intercity
11:03	1	Uitgeest	Sprinter
11:07	4	Rhenen	Sprinter
11:14	1	Schagen	Intercity

Relation of type $\mathit{Dates} \times \mathit{Times} \times \mathit{Platforms} \times \mathit{Stations} \times \mathit{TrainTypes}$

Binary relations and infix notation

A **binary relation** is a relation of type $A \times B$ or A^2
A relation **in the set A** is a binary relation of type A^2

Examples

$$\begin{array}{ll} \mathit{IsBrotherOf} := \{\langle x, y\rangle : x \text{ is brother of } y\} & \subseteq \mathit{People}^2 \\ \mathit{FollowsCourse} := \{\langle x, y\rangle : x \text{ follows } y\} & \subseteq \mathit{Students} \times \mathit{Courses} \end{array}$$

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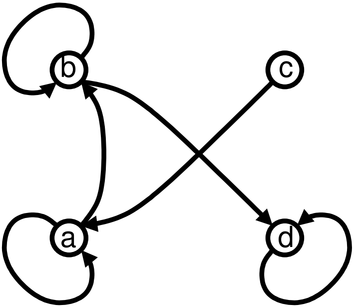
$$FollowsCourse := \{\langle x, y\rangle : x \text{ follows } y\} \subseteq Students \times Courses$$

Alternative infix notation

$x\ R\ y$	instead of	$\langle x, y\rangle \in R$
$x\ IsBrotherOf\ y$	instead of	$\langle x, y\rangle \in IsBrotherOf$
$x\ FollowsCourse\ y$	instead of	$\langle x, y\rangle \in FollowsCourse$

Visualization by directed graphs

Directed graphs are used to visualize relations in a set A:



$R = \{\langle a, a\rangle, \langle b, b\rangle, \langle d, d\rangle, \langle a, b\rangle, \langle b, d\rangle, \langle c, a\rangle\} \subseteq \{a, b, c, d\}^2$

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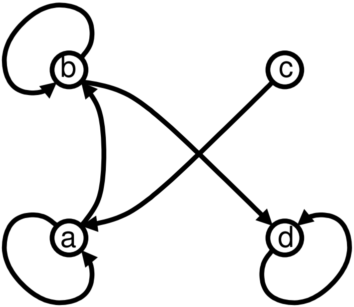
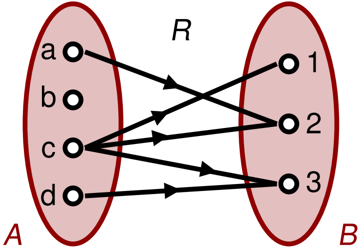


Matrix representation:

$$
\begin{array}{cc}
 & \begin{array}{cccc} a & b & c & d \end{array} \;\; \text{(to)} \\
\begin{array}{c} a \\ b \\ c \\ d \end{array} & \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
\text{(from)} &
\end{array}
$$

$R = \{\langle a,a\rangle, \langle b,b\rangle, \langle d,d\rangle, \langle a,b\rangle, \langle b,d\rangle, \langle c,a\rangle\} \subseteq \{a,b,c,d\}^2$

Venn diagrams can be used to visualize all binary relations:



$R = \{\langle a, 2\rangle, \langle c, 1\rangle, \langle c, 2\rangle, \langle c, 3\rangle, \langle d, 3\rangle\} \subseteq \{a, b, c, d\} \times \{1, 2, 3\}$

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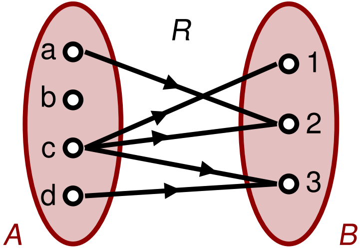

Matrix representation:

$$\begin{array}{c c} & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{array}{c} a \\ b \\ c \\ d \end{array} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

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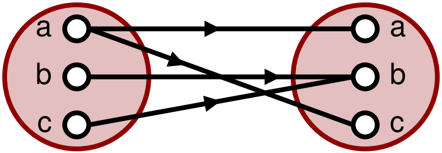
Exercise

The following Venn diagram depicts a relation R of type $\{a, b, c\}^2$:



Represent R by a directed graph, a matrix, and as a set of pairs.

break

Inverse of a binary relation

The **inverse** of a relation R is $\quad R^{-1} := \{\langle x, y\rangle : \langle y, x\rangle \in R\}$

$$\textbf{NB:} \quad R \subseteq A \times B \quad \Rightarrow \quad R^{-1} \subseteq B \times A$$

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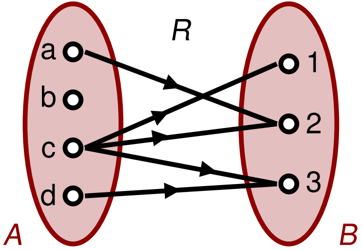
Example

LivesIn	$(\mathit{LivesIn})^{-1} = \mathit{IsResidenceOf}$
has type $\mathit{People} \times \mathit{Cities}$	has type $\mathit{Cities} \times \mathit{People}$
$\langle \mathit{Jan}, \mathit{Amsterdam}\rangle \in \mathit{LivesIn}$	$\langle \mathit{Amsterdam}, \mathit{Jan}\rangle \in (\mathit{LivesIn})^{-1}$
Jan LivesIn Amsterdam	*Amsterdam IsResidenceOf Jan*

The inverse in different representations

Inverting a relation $=$ reversing the arrows:



Matrix representation:

$$\begin{array}{c c} & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{array}{c} a \\ b \\ c \\ d \end{array} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

$$R \quad = \{\langle a,2\rangle, \langle c,1\rangle, \langle c,2\rangle, \langle c,3\rangle, \langle d,3\rangle\} \subseteq \{a,b,c,d\} \times \{1,2,3\}$$

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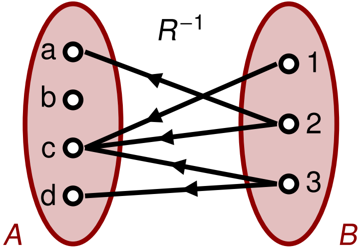

Matrix representation:

$$\begin{array}{c c} & \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array}$$

(transposed matrix)

$$R^{-1} = \{\langle 2,a\rangle, \langle 1,c\rangle, \langle 2,c\rangle, \langle 3,c\rangle, \langle 3,d\rangle\} \subseteq \{1,2,3\} \times \{a,b,c,d\}$$

Composition of binary relations

The **composition** of R and S (i.e. R **after** S) is the relation

$$R \circ S := \{\langle x, z\rangle : x\, S\, y \text{ and } y\, R\, z \text{ for some } y\}$$

NB: $S \subseteq A \times B,\ R \subseteq B \times C \quad \Rightarrow \quad R \circ S \subseteq A \times C$

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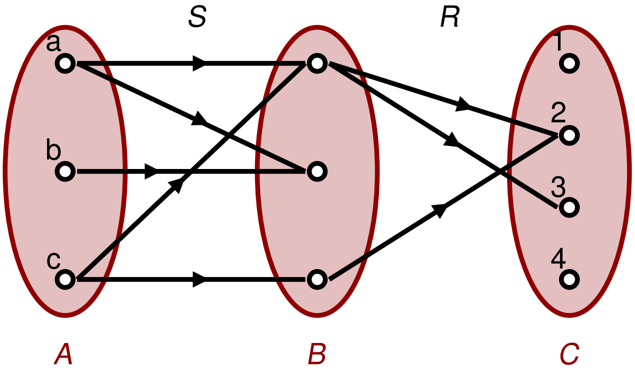
Example

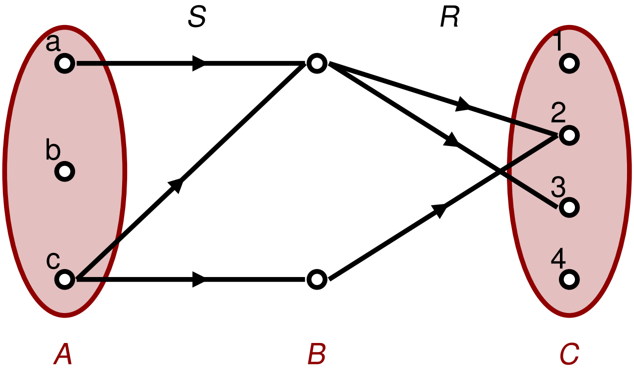
$R := \mathit{IsFriendOf} \circ \mathit{IsUncleOf}$

$\Rightarrow \quad R = \{\langle x, z\rangle : x \text{ is an uncle of a friend of } z\}$ **(note the the order!)**

$$\begin{tikzcd}
\text{John} \arrow[r, "\mathit{IsUncleOf}"] \arrow[rr, bend right, "\langle \text{John}, \text{Ann}\rangle \in R"'] & \text{Paul} \arrow[r, "\mathit{IsFriendOf}"] & \text{Ann}
\end{tikzcd}$$


S
R
a
b
c
1
2
3
4
A
B
C

Finding the composition using Venn diagrams

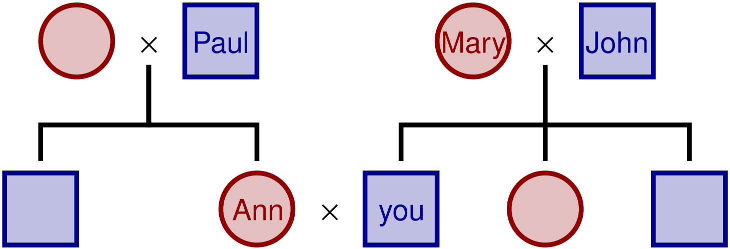


$$R \circ S = \{\langle a, 2\rangle, \langle a, 3\rangle, \langle c, 2\rangle, \langle c, 3\rangle\}$$

The order of composition matters:

$$\mathit{IsMarriedTo} \circ \mathit{IsParentOf} = \mathit{IsParentInLawOf}$$

$$\mathit{IsParentOf} \circ \mathit{IsMarriedTo} = \mathit{IsParentOf}$$



Exercise

In the set of *People*, consider the relations

$$IsParentOf := \{\langle x, y\rangle : x \text{ is parent of } y\}$$
$$IsSiblingOf := \{\langle x, y\rangle : x \text{ is brother or sister of } y\}$$

(a) What is the interpretation of the inverse of *IsParentOf* ?

(b) What is the interpretation of $(IsParentOf \circ IsParentOf)^{-1}$?

(c) What is the interpretation of $(IsParentOf \circ IsSiblingOf)^{-1}$?

Associativity of composition

Composition is associative: $(R \circ S) \circ T = R \circ (S \circ T)$

Example

Relation $\{\langle x, y \rangle : x$ is a son of a friend of an aunt of $y\}$

$$
\begin{tikzcd}
\text{John} \arrow[r, "\mathit{IsSonOf}", "T"'] \arrow[rr, bend left, "{\langle \text{John}, \text{Ann} \rangle \in S \circ T}"] & \text{Paul} \arrow[r, "\mathit{IsFriendOf}", "S"'] \arrow[rr, bend right, "{\langle \text{Paul}, \text{Betty} \rangle \in R \circ S}"'] & \text{Ann} \arrow[r, "\mathit{IsAuntOf}", "R"'] & \text{Betty}
\end{tikzcd}
$$

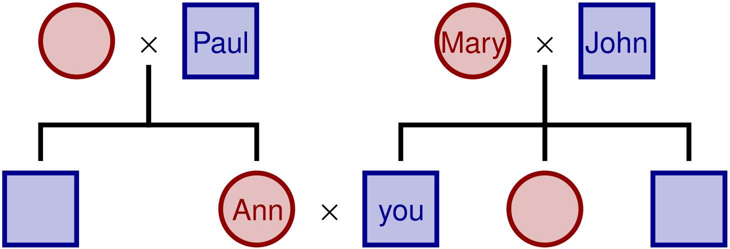
Inverse of a composite relation

The inverse of $R \circ S$ is $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$ **(note the order!)**

Example

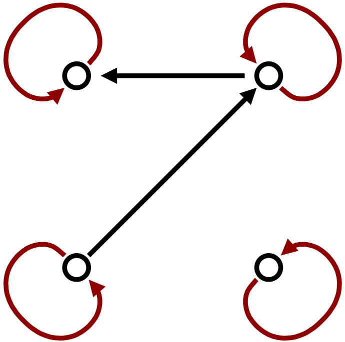
$$IsParentInLawOf = IsMarriedTo \circ IsParentOf$$

$$IsChildInLawOf = IsChildOf \circ IsMarriedTo$$



Properties specific for relations *in* a set A

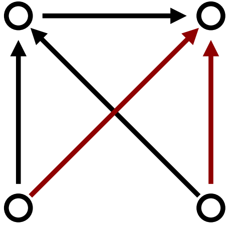
Reflexivity



for all elements x:

$x \, R \, x$

Transitivity

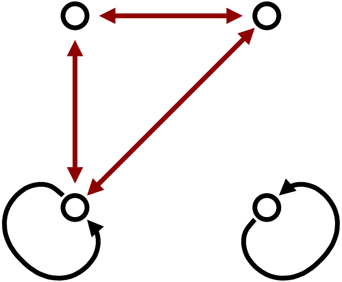


for all elements x, y, z:

$x \, R \, y \wedge y \, R \, z \rightarrow x \, R \, z$

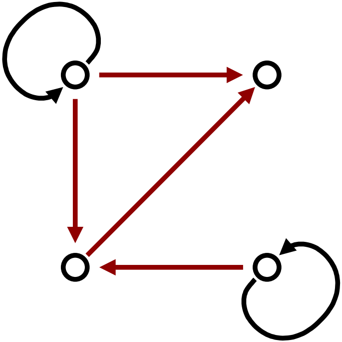
Properties specific for relations *in* a set A

Symmetry



for all elements x, y:
$x \, R \, y \rightarrow y \, R \, x$

Anti-symmetry



for all elements $x \neq y$:
$x \, R \, y \rightarrow \neg y \, R \, x$

Exercise

For each of the following relations in the set of *People*, strike through those properties that the relation does *not* (necessarily) have:

(a) *IsFriendOf*

reflexive transitive symmetric anti-symmetric

(b) *IsParentOf*

reflexive transitive symmetric anti-symmetric

(c) *IsDescendentOf*

reflexive transitive symmetric anti-symmetric